

APS110 Tutorial

#11

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Review – Thermodynamic Laws

1. All matter has a fundamental property called energy, which is defined by the fact that it is locally conserved

- $\Delta U_{\text{universe}} = \Delta U_{\text{system}} + \Delta U_{\text{surroundings}} = 0$
- For an isolated system: $\Delta U_{\text{system}} = 0$
- For a closed system: $\Delta U_{\text{system}} = q + w$

2. The entropy of the Universe increases during any spontaneous process (or stays the same)

- $\Delta S_{\text{universe}} = \Delta S_{\text{system}} + \Delta S_{\text{surroundings}} \geq 0$

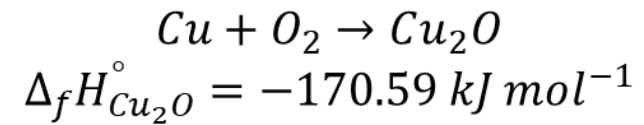
Review – Entropy

- In general, the Boltzmann entropy is $S = k_B \ln \Omega$, where Ω is the number of possible microscopic configurations that could result in the observed macroscopic state.
 - Systems with a high entropy have a high *potential* for randomness
 - Essentially, **entropy describes the uncertainty in knowing precisely the energy that an atom or molecule will have**
- For equilibrium systems, it can be shown that entropy can also be given by $\Delta S = \frac{q_{rev}}{T}$
 - Reversibility is an impossibly strict condition that states that all energy transfer occurs while the system remains in equilibrium. It works incredibly well for not actually being physically possible.

Review – Enthalpy of a Reaction

$$\Delta H_r = \Delta_f H^\circ_{products} - \Delta_f H^\circ_{reactants}$$

Example:



Review – Thermodynamic Variables

P: pressure, Pa

q: heat, J

R: ideal gas constant, 8.314 J/mol·K (or a bunch of other units if you need it)

T: temperature, K

U: internal energy, J

V: volume, m³

W: work, J

Review – Free Energy and Spontaneity

$$\Delta U = q + w$$
$$\Delta H = q = \Delta U + P\Delta V$$

At constant temperature, pressure:

$$\Delta S_{system} + \Delta S_{surroundings} > 0$$

$$\Delta S_{sys} - \frac{q_{rev}}{T} > 0$$

$$T\Delta S_{sys} - q > 0$$

$$T\Delta S_{sys} - \Delta H_{sys} > 0$$

$$\Delta G = \Delta H - T\Delta S$$

A free energy less than 0 means the process is spontaneous for the given reaction conditions
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Practice Question

1.1. If a 50 g ice cube at 0°C is added to 500 mL of water at 20°C, what is the equilibrium temperature of the system? Assume all heat transfer remains within the system.

You are provided the following information:

$$\rho_{\text{water}} = 1.00 \text{ g/mL}$$

$$c_{p, \text{ice}} = 2.06 \text{ J/g}^\circ\text{C}$$

$$c_{p, \text{water}} = 4.184 \text{ J/g}^\circ\text{C}$$

$$\Delta H_{f, \text{water}} = 6.01 \text{ kJ/mol}$$

Note: $q = mc\Delta T$, or $q = n\Delta H_f$ for phase change

Practice Question

Step 1) Ice melts

$$q_{ice(s)} = -q_{water}$$

$$q_{ice(s)} = n\Delta H_f = (50g) \left(\frac{mol}{18.02g} \right) \left(\frac{6.01kJ}{mol} \right) = 16.7 kJ$$

$$q_{water} = mc\Delta T \Rightarrow -16700 = (500mL) \left(\frac{1.00g}{mL} \right) \left(\frac{4.18J}{g^{\circ}C} \right) \Delta T$$

$$\Delta T = -7.98^{\circ}C \Rightarrow T_2 = T_1 - 7.98 = 12.02^{\circ}C$$

Practice Question

Step 2) Water formerly known as ice heats up

$$q_{ice(l)} = mc\Delta T = (50g) \left(\frac{4.18J}{g^{\circ}C} \right) (T_2 - 0^{\circ}C) = 205.2T_2$$

$$q_{water} = mc\Delta T = (500g) \left(\frac{4.18J}{g^{\circ}C} \right) (T_2 - 12.02^{\circ}C) = 2092T_2 - 25120$$

$$\begin{aligned} q_{ice(g)} &= -q_{water} \\ 205.2T_2 &= -[2092T_2 - 25120] \\ 205.2T_2 + 2092T_2 &= 25120 \\ T_2 &= 10.9^{\circ}C \end{aligned}$$

Practice Quiz

1. Which law of thermodynamics states that the entropy of a system and its surroundings never decreases, it either increases or stays the same?
 - a) Zeroth
 - b) First
 - c) Second
 - d) Third
2. Which of the following is always positive for a spontaneous reaction?
 - a) ΔS_{system}
 - b) $\Delta S_{\text{surroundings}}$
 - c) $\Delta S_{\text{universe}}$
 - d) $\Delta U_{\text{universe}}$

Practice Quiz

3. Propane is a commonly used fuel for portable barbeques.
- a) Write the balanced chemical equation for the combustion of 1 mol of gaseous propane (C_3H_8) in oxygen to form carbon dioxide gas and water vapour.
 - b) Determine the standard combustion enthalpy for one mol of gaseous propane
 - c) Assuming all of the heat from the combustion goes to heating 1 kg of water at 25°C , how many grams of water will boil off (form gas)?

You are provided the following information:

$$c_p = 4.184 \text{ J/g}\cdot\text{K}$$

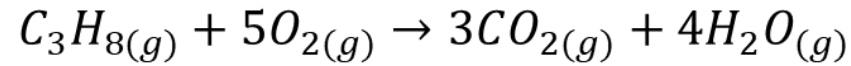
$$\Delta_{\text{vap}}H^\circ [\text{H}_2\text{O}] = 40.7 \text{ kJ/mol}\cdot\text{K}$$

$$\Delta_fH^\circ [\text{CO}_{2(g)}] = -393.5 \text{ kJ/mol}$$

$$\Delta_fH^\circ [\text{H}_2\text{O}_{(g)}] = -241.8 \text{ kJ/mol}$$

$$\Delta_fH^\circ [\text{C}_3\text{H}_{8(g)}] = -103.8 \text{ kJ/mol}$$

33a)



b)

$$\begin{aligned}\Delta_{comb}H^\circ &= 3\Delta_fH^\circ_{CO_{2(g)}} + 4\Delta_fH^\circ_{H_2O_{(g)}} - 1\Delta_fH^\circ_{C_3H_{8(g)}} \\ &= [3(-393) + 4(-241.8)] - [1(-103.8)] \\ &= -2040 \text{ kJ}\end{aligned}$$

c) Note: Masses are different (all water heats up to 100°C but not all water evaporates)

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$$\Delta_{comb}H^\circ = m_{heat}c\Delta T + (\Delta_{vap}H^\circ) \frac{m_{boil}}{4}$$

$$m_{boil} = \left[\frac{\Delta_{comb}H^\circ - m_{heat}c\Delta T}{\Delta_{vap}H^\circ} \right] \cdot 4$$

$$m_{boil} = \left[\frac{-(2040e3) - (1000)(4.184)(75)}{40.7e3} \right] \cdot 18.02$$

$$m_{boil} = 763 \text{ g}$$

Practice Quiz

4. Calculate the standard formation enthalpy for calcium carbide (CaC_2) reacting with water to form acetylene (C_2H_2) and calcium hydroxide ($\text{Ca}(\text{OH})_2$).

You are provided the following information:

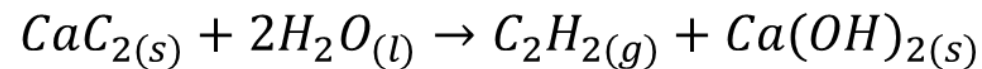
$$\Delta_f H^\circ [\text{C}_2\text{H}_2 (g)] = -83.9 \text{ kJ/mol}$$

$$\Delta_f H^\circ [\text{Ca}(\text{OH})_2 (g)] = -987 \text{ kJ/mol}$$

$$\Delta_f H^\circ [\text{H}_2\text{O} (l)] = -285.2 \text{ kJ/mol}$$

$$\Delta_f H^\circ [\text{CaC}_2 (g)] = -59.8 \text{ kJ/mol}$$

44.



$$\Delta_f H^\circ = 1\Delta_f H^\circ_{\text{C}_2\text{H}_{2(g)}} + 1\Delta_f H^\circ_{\text{Ca}(\text{OH})_{2(s)}} - [2\Delta_f H^\circ_{\text{H}_2\text{O}_{(l)}} + 1\Delta_f H^\circ_{\text{CaC}_{2(s)}}]$$

$$= [1(-83.9) + 1(-987)] - [2(-285.2) + 1(-59.8)]$$

$$= -439.5 \text{ kJ}$$

Additional Stuff

Quick entropy overview:

<https://opentextbc.ca/introductorychemistry/chapter/entropy-and-the-second-law-of-thermodynamics/>

A bit harder, but explains why $\Delta S = \frac{q_{rev}}{T}$:
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<https://galileo.phys.virginia.edu/classes/152.mf1i.spring02/Entropy.htm>
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